Department of Computer Engineering

Course Title	Image Processing												
Course Code	O23RC22												
Scheme	Learning Scheme						Assessment Scheme						
	L	T	P	SLH	NLH	Credits	# TS	TH	Total		SLA		Total
							Max	Max	Max	Min	Max	Min	Max
	4	-	-	-	8	4	30	70	100	40	-	-	100
Course Category	Core Technology Compulsory						Semester				6		

Indian Knowledge System = Nil

Abbreviations:	**TS – Test, TW – Term work, TH –** Theory Examination, **SLH** – Self Learning hours, **SLA** – Self Learning Assessment, NLH – Notional Learning Hours, IKS - Indian Knowledge System,
Legends:	# – Internal progressive Test assessment, @ – Internal progressive Term work assessment, $$ – online assessment, ** – External assessment,
Course Category:	S – Sciences & Humanities Course, B – Basic Technology Course, A – Allied Technology Course, C – Core Technology Course, M – Management Course, N – Non Technical course, X – Experiential Learning course, I – Institute Level Course

Course Outcomes

The Students will be able to

1. Explain the fundamentals of DTS and image processing
2. Apply image enhancement techniques to improve image quality.
3. Evaluate and apply image restoration and color image processing techniques
4. Analyze and implement image compression and segmentation approaches

Detailed Course Contents:

MARKS – 70

Sr. No.	Topic Name:	Marks	Hrs	CO	PO	PSO
1.	**Discrete Time Signals and Systems** - Introduction to digital signals and systems. - Properties and operations on digital signals. - Introduction to biosignals:-ECG,EEG,EMG,PPG - Classification of Signals, Systems, LTI System.	08	06	1	1,2,5	1,2,3
2.	**Introduction to Image Processing.** - What is image processing - Different types of images - Image : Pixel, Distance measures - Color fundamentals & models – RGB , HIS, YIQ - Basic steps in image processing - Image file formats : JPEG , PNG, GIF, TIFF, BMP - Operations on images: image addition, subtraction, logical operations, scaling, translation, rotation.	12	12	1	1,2,5	1,2,3
3.	**Image Enhancement** - Image Sensing and Acquisition, Image Sampling and Quantization, - Basic Relationships between Pixels: Neighbors of a Pixel, Adjacency and Connectivity, Distance Measures, Regions and Boundaries - Basic Gray Level Transformations (Linear, Logarithmic). - Spatial domain enhancement - Point Processing: Image Negative, Log Transformation - Contrast Enhancement:-Histogram, Histogram Equalization	20	18	2	1,2,3,4,5	1,2,3

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	- Filtering operations- Image smoothing, Image sharpening - Applications of Image Enhancement.				
4.	**Image Restoration and Color image processing**	15	15	3	1,2,3,4,5,6
	- Introduction to Image Restoration - Image Degradation Model - Difference between image enhancement and restoration - Mean Filters : Overview of Arithmetic Mean Filter, Geometric Mean Filter, Harmonic Mean F Color transformation: RGB → Grayscale, RGB → YCbCr, RGB → HSV, smoothing, sharpening a edge detection in color images.				
5.	**Image Compression and Image Segmentation**	15	14	4	1,2,3,4,5,7
	- Types of redundancy: Spatial Redundancy, Spectral Redundancy, Temporal Redundancy - Lossless compression: Run length coding, Huffman coding - Image Segmentation: Comparison of Point Detections, Line detection and Edge Detection - Video Compression Standards – JPEG, MPEG				

References:

Internet references – Audio visuals – Related Magazines – E-Journals:

Sr. No.	Topic Name	URL / Web address
1.	Introduction to Image Processing	https://nptel.ac.in/courses/117105135
2.	Image Enhancement	https://nptel.ac.in/courses/117105135
3.	Image Compression and Image Segmentation	https://nptel.ac.in/courses/117105135
4.	Image Restoration and Color image processing	https://nptel.ac.in/courses/117105135 https://www.tutorialspoint.com/dip/index.htm

Reference Books:

Sr. No.	*Title*	*Author*	*Publication*
1	Digital Image Processing	Rafael C. Gonzalez	GLOBAL EDITION
2.	Digital Image Processing: A Beginners' Guide	Dr. Gunasekaran M	Notion Press

GUIDELINES ON IMPLEMENTATION STRATEGIES

Theory course:

Topic wise tools used for teaching and learning

Ex. PPT, ICT, Handouts, Videos and Audio Visuals, Website Information

Sr. No	Topic Name	Tools used
1.	Introduction to Image Processing	PowerPoint Presentation, Google Classroom YouTube
2.	Image Enhancement	PowerPoint Presentation, Google Classroom YouTube
3.	Image Compression and Image Segmentation	PowerPoint Presentation, Google Classroom YouTube
4.	Image Restoration and Color image processing	PowerPoint Presentation, Google Classroom YouTube

Course Assessment and Attainment Strategies

a) # Progressive Test : 30 Marks 90 minutes

b) Theory Exam : 70 Marks 3 Hrs duration

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a) # Progressive Test (30 Marks)

The content is specified in terms of [30% (Assignment) + 30% (Test)] marks.

Assignment:

Assignment (on the provided answer sheets by the Institute) of 30 marks based on closed/ open note book/ sheet (A4 size) in own handwriting of student.

(The pattern of question paper shall be declared by the concerned faculty in the class.)

Test:

One test of 30 marks is conducted at departmental level. The format for the question paper is as per the norms laid down by department.

b) Theory Exam:

70 marks paper – 3 hrs.

Theory marks is conducted at institutional level by controller of examination at Exam Section. The format for the question paper is as per the norms laid down by exam section.

CO – PO – PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	2	-	-	1	-	-	3	2	3
CO2	2	3	2	2	3	-	-	3	2	3
CO3	2	3	3	2	3	-	1	3	2	3
CO4	2	3	2	3	3	-	1	3	2	3

Note:

1. **If a candidate does not satisfy the criteria of minimum attendance and / or the submission of Term –work of any course then the candidate shall be declared as "Detained" in that semester.**
2. **If a candidate does NOT secure the minimum passing marks in SLA of any course then the candidate shall be declared as "Fail" and will have to repeat and resubmit SLA work.**

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